Diagnosing Human-to-Robot Transfer Failures with a Transfer Gap Budget in Simulation

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Abstract: A failed rollout does not show which stage of human-to-robot transfer failed. We introduce the *Transfer Gap Budget*, which replaces one pipeline component at a time with a declared oracle and measures the paired change in task success. A joint oracle measures non-additivity. In an injected-error study with 500 fresh seeds per arm, perception has the largest budget on both the UR5e and UR10e. A look-again correction, selected on development seeds, raises success by 26.8 and 25.8 pp, respectively, the largest observed gains among four corrections. In exploratory runs the largest budget follows the assumed error model: perception under the nominal model, timing when speeds are drawn from the demonstrations (2–9$\times$ robot pace), calibration when calibration error is doubled. The budget-selected correction beats always applying look-again by 12–44 pp (95% intervals above 7 pp). On six demonstrations, diagnosis and a revised retreat raise paired success from 0–31/60 to 28–60/60 on unused seeds. Simulated depth perception meets pre-declared criteria on three manipulators. A G1 humanoid study finds 11/40 versus 31/40 placements for the same closed-loop controller across two runtime setups, with the cause unresolved. All robot evaluations are in simulation.

Keywords: human-to-robot transfer, failure diagnosis, oracle substitution

1 Introduction

Human-to-robot transfer requires motion estimation, retargeting, calibration, and executable timing. Aggregate success does not say which component to revisit, particularly with diverse human data [1]. We ask how much success correcting one pipeline component could recover. The *Transfer Gap Budget* measures each component's net oracle gain; gains can overlap. The largest development-set gain selects a correction for fresh evaluation.

2 Related Work

Human-to-robot transfer. H2RBench reports transfer gains by execution stage on a Franka arm [2]; we intervene on pipeline components. Human-as-Humanoid and HumanEgo learn from human video [3, 4]; hand-pose quality and motion mismatch affect cotraining [5, 6]. **Diagnosis and correction.** AHA, RoboFAC, and Guardian diagnose manipulation failures [7, 8, 9]. RHO and GaP search policy code and skill graphs [10, 11]. We use oracle gains to select from a fixed correction menu.

3 Method: the Transfer Gap Budget

Paired interventions. Let $\mathcal{K}$ contain perception, retargeting, tool-frame calibration, and execution timing. In the controlled-error study, *full* includes all four perturbations; $oracle_k$ removes only perturbation k, and *oracle-all* removes all four. Their ideal outputs are exact object pose, unperturbed

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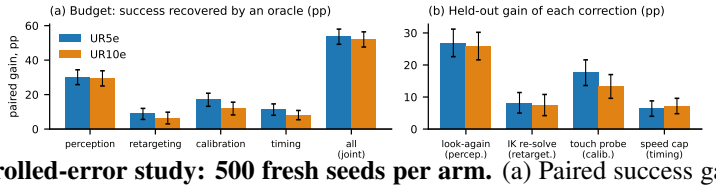


Figure 1: **Controlled-error study: 500 fresh seeds per arm.** (a) Paired success gains from single and joint oracles. (b) Gains from the four corrections; look-again was selected on development seeds. Bars: paired bootstrap 95% intervals.

IK, zero tool offset, and nominal timing. Each random stream is keyed by seed and component, preserving initial scenes and unreplaced error draws across conditions. For binary success $s_c(i)$ on seed i in condition c,

$$B_k = \frac{1}{n}\sum_{i=1}^{n}\big(s_{\text{oracle}_k}(i) - s_{\text{full}}(i)\big), \qquad \Delta_{\text{int}} = \sum_{k\in\mathcal{K}} B_k - B_{\text{all}}. \tag{1}$$

Here B_{all} uses the joint oracle in the same paired difference. Budgets are signed net gains: an oracle can both rescue and lose trials. The residual shows whether summed single-stage gains exceed (> 0) or fall below (< 0) the joint gain; it does not allocate failures to interactions.

Correction selection. The menu pairs perception with look-again, retargeting with an IK re-solve, calibration with a modelled touch probe, and timing with a speed cap. We select $k^\star = \arg\max_k B_k$ on development seeds, breaking ties in pipeline order. Look-again and probing reduce assumed residual errors; their manoeuvres are not simulated.

Statistics. Intervals are paired percentile bootstraps over seeds (95%), preserving each seed's condition scores. Demo intervals are descriptive: seeds 1000–1059 informed the revision.

4 Experimental Setup

Task. MuJoCo simulates a fixed-base UR5e or UR10e with a Robotiq 2F-85 from MuJoCo Menagerie [12], a 4 cm block with ±3 cm position jitter, and a 14 cm tray. A scripted IK controller uses simulated contact in a UR5e-tuned layout. Success requires a lift of at least 5 cm, release within 3 cm of the goal, and speed below 0.02 m/s for 0.5 s after release.

A: controlled errors. The injected magnitudes are assumptions. Both arms receive the same seed-dependent draws: Gaussian xy perception error ($\sigma = 1.5$ cm), joint offsets ($\sigma = 0.5°$ per joint), a uniformly directed tool-frame offset with magnitude $\mathcal{U}(0, 3)$ cm, and a timing speed-up of $\mathcal{U}(1.0, 4.5)$ relative to robot pace. The corrections reduce perception noise to 0.4 cm, joint-offset noise to $0.05°$, or calibration noise to 2 mm per axis, or cap speed-up at $3\times$. Development seeds 0–59 select look-again on both arms. After freezing the protocol, we evaluate ten conditions on 500 fresh seeds (3000–3499) per arm: *full*, four single oracles, the joint oracle, and four corrections.

E: sensitivity. On further fresh seeds (200 per arm unless stated) we repeat A's conditions with errors scaled by 0.5 (500 seeds) and 2, a 3 or 5 cm block, ±5 cm jitter, the tray or block moved 8–10 cm, perception error scaled 0.1–2, cut to 25% alone or with calibration error doubled (500 seeds), set to the depth estimator's measured error, and timing drawn from the demonstrations' measured speeds (500 seeds). Correction constants are unchanged.

B: real-video motion. Five pick-and-place cycles from two people in two freely licensed videos, plus one from a third person's 50 s recording, supply motion and timing. GEM-X [13] estimates hand trajectories; grasp and release frames are annotated manually. Reach and retreat are anchored to the simulated grasp and release points, and the carry path is rotated and scaled to connect them; the scene is not reconstructed. Perception and calibration errors remain injected; no object detector is used, and retargeting is excluded from the real-video oracle analysis. Each demo–arm pair uses 60 seeds from the revision period (1000–1059) and 60 unused seeds (5000–5059). We report leg-wise path RMSE against the retargeted reference.

C: depth perception. A geometric estimator fits the table plane, clusters points above it, and selects a cube and tray; its position, yaw, and size drive the controller. We test 5, 6, and 7 cm cubes on the two arms and a third manipulator (UR10e with a Robotiq 2F-140), with 40 seeds per size: 120 episodes per condition. **D: humanoid.** A Unitree G1 with Dex3 hands uses SONIC whole-body control [14]. We score grasp (5 cm lift with contact) and placement within 5 cm of the goal; we do not estimate stage budgets here.

5 Results

5.1 Controlled errors on two arms

Uncorrected success is 232/500 on the UR5e and 240/500 on the UR10e. Removing all errors recovers 53.6 and 52.0 pp. Perception has the largest budget: 30.0 [25.8, 34.4] pp on the UR5e and 29.4 [25.0, 33.8] pp on the UR10e. Calibration follows (17.0 and 12.0 pp), then timing (11.2 and 8.0 pp) and retargeting (8.8 and 6.4 pp); all single-stage intervals exclude zero (Fig. 1a).

The interaction residual is 13.4 [6.2, 20.8] pp on the UR5e and 3.8 [-2.6, 10.4] pp on the UR10e.

Look-again, selected on development trials, raises success from 232 to 366/500 (UR5e) and 240 to 369/500 (UR10e); the paired gains are 26.8 [22.6, 31.2] pp and 25.8 [21.6, 30.2] pp, respectively, the largest observed. Probing gives 17.6 and 13.4 pp; IK re-solving and the speed cap 6–8 pp (Fig. 1b).

5.2 Sensitivity to assumptions, layout and the budget rule

Perception has the largest budget at nominal and doubled errors, with 3 or 5 cm blocks, $\pm$5 cm jitter (ten cells), and moved tray or block (six cells). At $0.5\times$ errors (484 and 485 of 500 uncorrected) it still leads, by 0.8 pp [0.2, 1.6] over calibration on both arms. Varying only perception error (Fig. 2), perception leads from the assumed 1.5 cm upward; at $0.75\times$ and below calibration and timing take over.

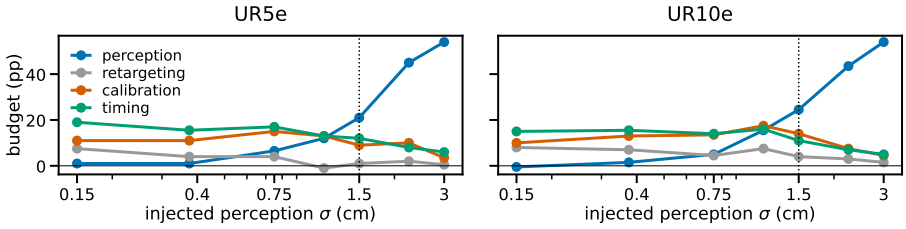


Figure 2: Budgets against injected perception error (200 seeds per point); dotted: assumed value.

Tying magnitudes to measurements changes the ranking. The depth estimator's block-position error is 0.34 mm RMS per axis with exact camera calibration, and 7.1 and 14.2 mm with $0.5°$/5 mm and $1°$/10 mm calibration error. Injected as the perception error, these give perception budgets of -1.0 and 2.0 pp, 10.0 and 9.5 pp (timing leads at both), and 28.5 and 31.0 pp (perception leads). The six demonstrations run at 2.0–9.4$\times$ robot pace, above the assumed 1–4.5$\times$. Drawing speed from that range (72 and 87 of 500 uncorrected), timing has the largest budget, 50.2 and 44.2 pp, 38.6 [32.6, 44.2] and 35.8 [30.0, 41.4] pp above the next stage. Calibration and retargeting magnitudes remain assumed.

With 500 seeds per arm, the largest budget on the first 100 selects a correction evaluated on the other 400. The rule picks timing with perception error cut to 25% (gains 14.0 and 13.0 pp against 2.2 and 1.0 pp for look-again: +11.8 [7.5, 16.0] and +12.0 [7.8, 16.2] pp; calibration was best in hindsight, 1.2 and 1.5 pp short, intervals include zero), calibration when calibration error is also doubled (43.8 and 44.5 pp against about zero: +43.8 [39.0, 48.5] and +44.0 [39.0, 49.0] pp; equal to the best), and timing at measured speeds (42.5 and 44.0 pp: +31.8 [25.0, 38.2] and +35.2 [28.7, 41.8] pp; equal to the best). Where perception dominates it picks look-again and ties with that fixed choice.

5.3 Real-video demonstrations

Table 1: Successes of 60 paired seeds, uncorrected → revised stack; left seeds were reused in development, right seeds not. P1–P3: demonstrators.

	Seeds 1000–1059		Unused seeds	
Demonstration	UR5e	UR10e	UR5e	UR10e
P1 notebook	0→ 57	0→ 60	0→ 60	0→ 60
P1 markers	0→ 51	0→ 25	0→ 49	0→ 28
P1 orange marker	0→ 46	0→ 58	0→ 51	0→ 59
P1 notepad	0→ 58	0→ 58	0→ 60	0→ 60
P2 box (own stack)	11→ 60	4→ 55	23→ 60	9→ 49
P2 box (P1 stack)	11→ 60	4→ 50	23→ 60	9→ 52
P3 recording (P1 stack)	35→ 59	30→ 56	31→ 57	21→ 53

The notebook demo fails uncorrected on both arms at about $6\times$ robot pace. Sequential diagnosis selects a $3\times$ speed cap, look-again, calibration probing, then a tighter $2\times$ cap; with the revised retreat the stack reaches 57/60 and 60/60 (Table 1), and RMSE to the retargeted reference falls from 13.9 to 4.9 cm (UR5e) and 9.1 to 2.9 cm (UR10e).

The original stack scored 0/60 on all 3 additional P1 cycles, whose copied retreat paths intersect the tray, and cut P2 success on the UR5e from 11/60 to 0/60. We therefore added a 5 cm lift before retreating along the human path (selected on development seeds 0–19; path-RMSE cost 2.7–4.1 cm). The revised P1 stack transfers to P2 and P3 without retuning; P1 markers remain difficult on the UR10e (25/60). Seeds 1000–1059 informed the revision, so those columns are exploratory. On seeds 5000–5059, which no development step used, every cell improves: the revised stack reaches 28–60/60 against 31 or fewer uncorrected, all paired intervals exclude zero, and P1 markers on the UR10e reaches 28/60.

5.4 Depth perception and humanoid runtime sensitivity

All 30/30 pre-declared acceptance cells reach the 90% point-estimate threshold (113–120 of 120 episodes), covering yaw, distractors, depth noise, occlusion with a retry, and combined stressors on three manipulators. Report-only stress tests expose limits: 2°/20 mm calibration error gives 25/120 on the UR5e, and occlusion without retry 0–16/120. For the notebook path at robot pace, depth versus ground-truth success is 109 versus 108/120 (UR5e), 112 versus 111/120 (UR10e), and 120 versus 118/120 (third arm): 0.8–1.7 pp higher, with path RMSE within 0.01 cm. All three meet the point-estimate margins of 10 pp success loss and 0.5 cm added RMSE (no inferential non-inferiority test).

G1 humanoid. With IK-planned targets, the same frozen closed-loop controller places 11/40 objects in a volume-based installation and 31/40 in a packaged container (same GPU, seeds 1200–1239; discordant placements 20 to 0, exact McNemar $p = 1.9 \times 10^{-6}$). Code and environment checks did not identify the cause; a four-seed MuJoCo 3.15.0 rebuild (0/4 lifts versus 4/4 on 3.14.0) also changed `filelock`.

6 Limitations and Conclusion

The study uses one simulated task, scripted controllers, a UR5e-tuned layout, and assumed error and correction magnitudes; only the perception and timing magnitudes are tied to measurements, and layout runs move the tray and block, not the table or task. Six demonstrations from three people and unreconstructed scenes limit generalisation; look-again and probing cost no simulated time. Perception has the largest budget under the assumed model; with demonstration-measured speeds timing does, and the budget-selected correction follows. This supports diagnosis in simulation; real-robot performance remains untested. An AI assistant helped draft parts of the manuscript and code; values were checked against saved results.

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